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Profiling the plight of adolescent girls and young women: risks and pathways for intervention in rural Uganda

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Abstract

Background Adolescent girls and young women (AGYW) in rural Uganda face overlapping socio-economic, educational, and behavioral vulnerabilities that heighten their risks of HIV, unintended pregnancies, gender-based violence (GBV), and other adverse health outcomes. Establishing baseline risk profiles is essential for designing targeted interventions.

Methods This study analyzed secondary data collected through the Uganda DREAMS and OVC Tracking System (UDOTS) during program enrollment between July 2017 and December 2023 in four districts of Central Uganda (Mubende, Mityana, Luwero, and Kassanda). UDOTS screening tools captured socio-economic, educational, and behavioral indicators. K-means cluster analysis was applied to classify participants into High-Risk and Low-Risk groups, followed by bivariate validation. Logistic regression was used to identify predictors of high-risk classification.

Results Of the 124,259 AGYW screened, 24.7% were categorized as High-Risk. This group was predominantly aged 20–25, out of school, engaged in transactional sex, and exposed to violence. In contrast, the 9–14 age group largely clustered as Low-Risk, reflecting delayed sexual debut, stronger school retention, and lower exposure to economic pressures. Education was strongly protective: AGYW enrolled in school had substantially greater odds of being classified as Low-Risk (AOR = 108.15, 95% CI: 93.84–124.65, $p < 0.001$). Conversely, transactional sex strongly predicted High-Risk classification (AOR = 1599.97, 95% CI: 769.09–3328.47, $p < 0.001$).

Conclusion AGYW in rural Uganda particularly those aged 20–25, face compounded vulnerabilities linked to poverty, violence, and school dropout. Strengthening education, economic empowerment, and GBV prevention is critical to reducing these risks and fostering resilience.

Keywords Adolescent girls and young women, Risk profiling, Transactional sex, School dropout, Violence and vulnerability

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Introduction

Adolescent girls and young women (AGYW) in sub-Saharan Africa (SSA) face multiple social, economic, and cultural vulnerabilities that expose them to adverse health and social outcomes, including HIV, unintended pregnancies, and sexually transmitted infections (STIs). They are also disproportionately affected by gender-based violence (GBV), school dropout, and limited economic opportunities [1, 2]. In SSA, AGYW account for 87% of global HIV infections among women aged 15–24 [3, 4]. In Uganda, they contribute to more than two-thirds of new HIV infections, underscoring the urgency of addressing structural and behavioral drivers of vulnerability [5].

HIV risk among AGYW is shaped by a combination of socio-economic deprivation, early marriage, and low educational attainment, which often push young women into transactional sex with older partners who wield greater power in relationships [6]. Such power imbalances reduce their ability to negotiate safe practices, increasing the risk not only of HIV but also of other STIs and unintended pregnancies. Broader gender inequality entrenched in patriarchal norms further limits AGYW's autonomy over their sexual and reproductive health [7, 8].

Poverty is a consistent driver of vulnerability, as economic desperation forces many AGYW into survival strategies such as transactional sex, where their lack of bargaining power heightens risks of coercion and unprotected sex [1, 9]. The absence of strong social safety nets compounds these challenges, leaving AGYW dependent on financially dominant relationships for basic needs [10, 11].

Educational disengagement adds another layer of risk. Girls who drop out of school due to early pregnancy, caregiving responsibilities, or lack of school fees face diminished opportunities for advancement [12, 13]. Evidence shows that out-of-school AGYW are more likely to initiate sexual activity early and engage in risky behaviors compared to their peers who remain in school [6]. Conversely, school attendance has been consistently linked to delayed sexual debut and reduced likelihood of early marriage, confirming education's role as a strong protective factor [14].

GBV remains widespread, with more than 40% of AGYW in SSA reporting some form of physical, sexual, or emotional violence [15–17]. These experiences undermine mental health, limit access to care, and increase the likelihood of unsafe sexual behaviors [9, 18]. Substance use including alcohol, marijuana, and other illicit drugs further exacerbates these risks. Use of such substances is linked to impaired decision-making, unprotected sex, and multiple sexual partnerships, reinforcing cycles of vulnerability [7, 19].

To address these intersecting risks, the Determined, Resilient, Empowered, AIDS-free, Mentored, and Safe

(DREAMS) initiative was launched by the U.S. President's Emergency Plan for AIDS Relief (PEPFAR) in 2014. DREAMS operates in 10 SSA countries with the highest HIV burden, including Uganda, and delivers a layered package of interventions such as life skills education, economic empowerment, GBV prevention, and sexual and reproductive health services [20]. In Uganda, DREAMS is implemented in selected districts with documented HIV and GBV vulnerabilities, supported by para-social worker networks and safe-space programming. A key feature of the program is its risk stratification process, which uses standardized screening tools to categorize AGYW and tailor interventions to their specific needs [1].

While DREAMS has demonstrated promise in reducing HIV incidence, most evaluations emphasize program outcomes rather than the baseline vulnerabilities of participants. Yet, understanding the socio-economic, educational, and behavioral risks at enrollment is essential to address root causes and inform targeted, sustainable programming. Current literature often focuses on intervention impact while overlooking these baseline vulnerabilities, creating a critical knowledge gap [21].

This study addresses that gap by analyzing enrollment data from 124,258 AGYW in four rural districts of Central Uganda: Mubende, Mityana, Kassanda, and Luwero. The objectives were:

(a) to examine the socio-economic, educational, and behavioral vulnerabilities of AGYW, including household headship, employment status, school attendance, reasons for dropout, early pregnancy, transactional sex, and experiences of violence, and to analyze their associations with other risk factors; and (b) to identify and characterize distinct risk profiles of AGYW using cluster analysis, highlighting intersections of vulnerabilities for targeted intervention planning.

By profiling AGYW at baseline, this study moves beyond program outcome evaluation to highlight the underlying risks that shape their trajectories. Establishing such baselines not only strengthens monitoring and evaluation frameworks but also ensures that interventions and policies are evidence-based, context-specific, and responsive to the realities of AGYW in rural Uganda.

Study design

This study utilized a retrospective, cross-sectional design to analyze secondary data from 124,258 AGYW across four districts in Uganda. The dataset was drawn from the Uganda DREAMS and OVC Tracking System (UDOTS), which applied standardized screening tools during participant intake between July 2017 and December 2023. The DREAMS initiative was introduced in Uganda in 2015 as part of PEPFAR and is implemented in selected high-burden districts rather than nationwide. It seeks to reduce HIV risk and related vulnerabilities among

adolescent girls and young women by addressing socio-economic, educational, and behavioral risk factors.

The primary objective of this study was to leverage baseline enrollment data to establish risk profiles of AGYW at the point of entry into the program. Endline data were excluded because the study aimed to document vulnerabilities before exposure to DREAMS' layered interventions. This allowed us to capture the unmitigated socio-economic, educational, and behavioral risks faced by AGYW and provide a more accurate picture of their baseline conditions.

Study setting and participants

The DREAMS initiative in Uganda was implemented in four districts of Central Uganda: Mubende (a predominantly rural district with trading hubs and referral hospital catchments), Mityana (a peri-urban–rural mix with relatively denser health facility coverage), Luwero (agro-pastoral communities with dispersed settlements and lower school retention among AGYW), and Kassanda (a newer district with artisanal mining activities and mobile populations). Districts were purposefully selected because they had (i) existing DREAMS safe spaces and para-social worker infrastructure, (ii) documented HIV, GBV, and other vulnerabilities among AGYW, and (iii) operational feasibility within the CDC-funded Mubende Region HIV Project (Project Code 18033). Central Uganda overall has a higher HIV prevalence (6.2%) than the national average (5.8%), underscoring the importance of focusing on this region [20].

The dataset analyzed in this study was drawn from AGYW screened during enrollment. Inclusion criteria required participants to meet DREAMS risk thresholds and have complete screening data. Records with missing or duplicated entries were excluded. The analysis included 124,259 AGYW. Key outcome variables of interest were transactional sex, multiple sexual partnerships (MSP), and experiences of violence, alongside educational and socio-economic indicators.

Data collection

The data analyzed in this study were collected using age-specific standardized screening tools (for age groups 10–14, 15–19, and 20–24 years), specifically developed for the DREAMS Project. These tools were administered through one-on-one interviews conducted by trained social workers in private settings to ensure confidentiality and comfort for participants. The screening tools captured a comprehensive range of socio-economic, educational, and behavioral indicators across the following key domains:

Socio-economic indicators

Information was collected on household headship (e.g., whether participants lived in child-headed households),

work status (whether participants were engaged in income-generating activities), and living conditions, which reflected the overall socio-economic stability of the household.

Educational indicators

Data were collected on school attendance status (in school or out of school), the number of school days missed during the last term, reasons for absenteeism or dropout (such as lack of school fees or family responsibilities), and interest in returning to school for those who were out of school.

Sexual and reproductive health indicators

Participants provided information on their age at first pregnancy, history of sexual activity (including consensual, coerced, or forced experiences), engagement in transactional sex (exchanging sex for items or favors), and the number of sexual partners in the past 12 months.

Violence indicators

Participants were asked about their experiences with physical violence, emotional/psychological abuse, and sexual violence. These responses helped identify AGYWs who had been subjected to various forms of harm.

Substance use indicators

Data were gathered on alcohol consumption and illicit drug use, as well as any consequences of substance use, such as social or health problems. All data were entered into a centralized database, ensuring standardization and accuracy. The structured tool was designed to comprehensively capture the vulnerabilities of AGYWs at the time of their enrollment into the DREAMS program.

Alcohol and substance-use assessment. Screening questions were included in the DREAMS enrollment tool. These asked participants: “Have you consumed alcohol in the past 30 days?”, “Have you ever used illicit drugs such as marijuana, khat, or shisha?”, and “In the past 30 days, have you had more than three alcoholic drinks on one occasion or used drugs frequently?” These questions were adapted from validated survey modules (Uganda Demographic and Health Survey; WHO STEPS) but streamlined for program use. Responses were coded as binary indicators (Yes/No) for analysis.

Parental status Was measured by asking whether both biological parents were alive and co-residing with the participant; responses were coded “yes” (both alive and co-residing) or “no” (one or both deceased/not co-residing).

Data analysis

The dataset was cleaned, coded, and analyzed using SPSS Version 27 to explore the vulnerabilities of adolescent

girls and young women. Descriptive statistics provided an overview of key variables, with frequencies and percentages summarizing categorical data such as school attendance and household headship, while means and standard deviations were used for continuous variables like age. To classify participants into distinct risk groups, K-means clustering analysis was employed. This method effectively partitioned participants into High Risk and Low-Risk clusters based on variables such as household headship, work status, school attendance, transactional sex, experiences of violence, and age at first pregnancy, offering a nuanced understanding of the multidimensional vulnerabilities within the population.

Clustering approach and assumptions. We used K-means because it scales efficiently to very large samples and provides an interpretable partition for downstream validation. We note its distributional assumptions approximately spherical clusters, similar within-cluster variance may not fully hold for psychosocial indicators. To address this, we emphasized multi-method validation (cluster separation, external variable association, and regression-based prediction).

To validate the clusters, bivariate analysis was conducted, examining associations between cluster membership and key outcomes such as transactional sex, multiple sexual partners, and experiences of violence. Chi-square tests assessed these relationships, while odds ratios quantified the association's strength, confirming the clusters' distinctiveness. Model specification and variable selection. Variables entered into the adjusted logistic regression (school status, sexual activity, and transactional sex) were chosen a priori based on theory, strength and stability of bivariate associations, and collinearity screening. Several additional candidates e.g., parental status, employment were excluded to avoid redundancy and model instability given strong inter-correlations with included predictors and sparse cells in some strata. This parsimonious specification prioritizes interpretability while minimizing overfitting. Logistic regression models were subsequently used to identify predictors of high-risk behaviors, providing deeper insights into the socio-economic and behavioral factors shaping AGYW's risk profiles. This combined approach ensured the robustness and relevance of the clustering results, offering a solid foundation for targeted interventions.

Missing data and diagnostics. Missingness across predictors was <5% and showed no concerning patterns on inspection; we used complete-case analysis. We assessed multicollinearity (all VIF < 2.5; Condition Index < 5), examined influential observations, and checked small-cell issues by (i) collapsing rare categories where appropriate and (ii) re-estimating models with robust standard errors. In all cases, inferences were unchanged.

Ethical considerations

This study adhered to the ethical principles outlined in the Declaration of Helsinki and was conducted under the ethical approvals granted by the Mildmay Uganda Research and Ethics Committee (MUREC) (REC REF 0804–2018) and the Uganda National Council of Science and Technology (SS639ES). Permission to use the database for this secondary analysis was obtained through administrative clearance from Mildmay Uganda, ensuring compliance with institutional guidelines for data access and usage.

A waiver of assent and parental consent for the use of secondary data was granted by the Mildmay Research Ethics Committee (REC) due to the retrospective nature of the study. This waiver was deemed appropriate as the study utilized pre-existing, anonymized data, with no direct interaction with participants or collection of new data. All data extracted from the database were fully anonymized before analysis to safeguard participants' privacy. Identifying information was removed, ensuring confidentiality in both the analysis and reporting stages of the study.

Sensitivity and robustness checks

We conducted three sets of sensitivity analyses to evaluate the stability of estimates: [1] Alternative codings: We collapsed rare categories for violence and substance-use indicators and re-estimated the models; effect directions and significance persisted [2]. Augmented covariates: We added parental status and employment to the adjusted model to probe omitted-variable concerns; core predictors (school status, sexual activity, transactional sex) remained significant with similar magnitudes [3]. Influence and precision: We re-fit models with robust standard errors and inspected influence diagnostics; no single cluster or district unduly affected results. Collectively, these checks indicate that results are robust to reasonable alternative specifications.

Results

Background characteristics of study participants

The study included a total of 124,259 respondents, with a mean age of 14.57 years (SD = 4.369). The age distribution showed that 63.9% or 79,374 participants were aged 9–14 years, 22.7% or 28,200 participants were aged 15–19 years, and 13.4% (16,685) were aged 20–25 years. This reflects a diverse age composition across the groups, providing a comprehensive representation of the target population.

Cluster analysis and socio-economic risk profiling

The cluster analysis classified participants into two groups: a high-risk group consisting of 24.7% (30,663

participants) and a low-risk group consisting of 75.3% (93,595 participants). The separation between these groups, indicated by a Euclidean distance of 2.689, was statistically valid. Differences were significant across multiple domains. For transactional relationships, a larger proportion of the high-risk group reported having stayed in a relationship expecting gifts, favors, or financial help ($F = 85,902.655$, $p < 0.001$). School status also differed sharply, with most of the high-risk group being out of school compared to the majority of the low-risk group who were enrolled ($F = 105,585.313$, $p < 0.001$).

Table 1 ANOVA results for behavioral risk profiles of adolescent girls and young women (AGYW) in rural Uganda

Risk Question	F	P-value
Stayed in a relationship for gifts or favors?	85,902.655	0.000
Experienced forced or coerced sexual activity?	554,068.366	0.000
Alcohol or drugs caused problems in the last three months?	6,492.030	0.000
Currently sexually active?	500,465.289	0.000
Engaged in transactional sex?	73,207.282	0.000
Currently working?	10,313.093	0.000
Experienced sexual violence?	10,598.956	0.000
Experienced physical violence?	2,875.226	0.000
Threatened with a knife, gun, or weapon in the last year?	871.961	0.000
Told you were unloved or undeserving of love in the last year?	2,473.662	0.000
Told you were unloved, wished dead, or made to feel worthless?	1,590.401	0.000
Threatened with a knife, gun, or weapon in the past year?	1,402.222	0.000
Punched, kicked, whipped, or beaten with an object in the past year?	5,371.876	0.000
Used condoms regularly with the most recent partner?	10,253.498	0.000
Choked, smothered, or burned intentionally in the past year?	714.399	0.000
Experienced ridicule, insults, or emotional neglect?	8,098.577	0.000
Forced to have sex through physical force, harassment, threats, or tricks?	1,734.619	0.000
Inappropriately touched (e.g., fondling or touching sexual body parts)?	9,786.720	0.000
Had more than three drinks or used drugs frequently?	1,367.420	0.000
Consumed alcohol or used illicit drugs (e.g., marijuana, cocaine, shisha)?	9,463.589	0.000
Are both parents living?	19,841.502	0.000
Currently in school?	105,585.313	0.000
Number of sexual partners in the last 12 months?	335,632.865	0.000
Age group?	625,795.458	0.000
Is the head of the household under the age of 18?	8.520	0.004
Ever been pregnant?	5,739.747	0.000

Reports of violence and emotional abuse were also concentrated in the high-risk group. Participants in this group more frequently indicated experiences of sexual violence ($F = 10,598.956$, $p < 0.001$), physical violence ($F = 2,875.226$, $p < 0.001$), and continuous ridicule, insults, or emotional neglect ($F = 8,098.577$, $p < 0.001$). Substance use showed significant disparities, with the high-risk group reporting higher rates of alcohol or illicit drug use ($F = 9,463.589$, $p < 0.001$) as well as heavy or frequent use ($F = 1,367.420$, $p < 0.001$). Household headship also differed, with the high-risk group more frequently associated with households headed by individuals under the age of 18 ($F = 8.520$, $p = 0.004$). These findings demonstrate that the two-cluster model captured statistically distinct patterns across socio-economic, educational, and behavioral domains (Table 1).

Profiling risk distribution among adolescent girls and young women in rural Uganda: a cluster and bivariate analysis

The analysis followed a two-step approach: participants were first segmented into High-Risk and Low-Risk groups using cluster analysis, and the results were validated through bivariate analysis (Table 2). Significant differences were observed between the two groups across socio-economic, educational, and behavioral dimensions ($p < 0.001$ for all key variables).

Cluster composition and validation

Cluster analysis classified 24.7% ($n = 30,664$) of participants into the High-Risk group and 75.3% ($n = 93,594$) into the Low-Risk group. Age was a key differentiator, with 92.0% of High-Risk participants aged 20–24 years compared to 84.8% of Low-Risk participants aged 10–14 years ($p < 0.001$). The Euclidean distance between the two clusters was 2.689, indicating clear separation.

Socio-economic characteristics

Among High-Risk participants, 94.0% reported that both parents were not present, compared to 52.4% in the Low-Risk group ($p < 0.001$). Employment status differed significantly, with 9.9% of High-Risk participants engaged in income-generating activities compared to almost none in the Low-Risk group ($p < 0.001$).

Educational characteristics

School attendance varied sharply between groups. While 83.6% of Low-Risk participants were currently in school, 91.4% of High-Risk participants were not enrolled ($p < 0.001$).

Behavioral characteristics

Sexual activity was more prevalent among High-Risk participants: 85.3% reported being sexually active compared

Table 2 Risk distribution among adolescent girls and young women (AGYW) in rural Uganda: cluster and bivariate analysis

Variable	Category	High Risk (N, %)	Low Risk (N, %)	p-value
Age Group (years)	10–14	2 (0.0)	79,372 (84.8)	< 0.001
	15–19	2,462 (8.0)	14,223 (15.2)	
	20–24	28,199 (92.0)	0 (0.0)	
Alcohol/Drug Use	Yes	7,486 (24.4)	5,302 (5.7)	< 0.001
	No	23,177 (75.6)	88,293 (94.3)	
Parental Status	Both alive & co-residing	1,843 (6.0)	44,595 (47.6)	< 0.001
	One/both not present	28,820 (94.0)	49,000 (52.4)	
Household Head < 18	Yes	0 (0.0)	26 (0.0)	0.004
	No	30,663 (100.0)	93,569 (100.0)	
Education Status	In school	2,644 (8.6)	78,217 (83.6)	< 0.001
	Out of school	28,019 (91.4)	15,378 (16.4)	
Sexual Activity	Yes	26,153 (85.3)	296 (0.3)	< 0.001
	No	4,510 (14.7)	93,299 (99.7)	
Condom Use	Yes	3,081 (10.0)	34 (0.0)	< 0.001
	No	27,582 (90.0)	93,561 (100.0)	
Transactional Sex	Yes	13,470 (43.9)	9 (0.0)	< 0.001
	No	17,193 (56.1)	93,586 (100.0)	
Threatened with Weapon	Yes	1 (0.0)	4,093 (4.4)	< 0.001
	No	30,662 (100.0)	89,502 (95.6)	
Physical Violence	Yes	8,024 (26.2)	8,024 (8.6)	< 0.001
	No	22,639 (73.8)	85,571 (91.4)	
Sexual Violence	Yes	7,543 (24.6)	4,812 (5.1)	< 0.001
	No	23,120 (75.4)	88,783 (94.9)	
Employment Status	Yes	3,045 (9.9)	1 (0.0)	< 0.001
	No	27,618 (90.1)	93,594 (100.0)	
Frequent Alcohol/Drug Use	Yes	470 (1.5)	18 (0.0)	< 0.001
	No	30,193 (98.5)	93,577 (100.0)	

to 0.3% in the Low-Risk group ($p < 0.001$). Engagement in transactional sex was reported by 43.9% of High-Risk participants, compared to almost none in the Low-Risk group ($p < 0.001$). Alcohol or illicit drug use was reported by 24.4% of High-Risk participants compared to 5.7% of Low-Risk participants ($p < 0.001$). Frequent or heavy substance use was reported by 1.5% of High-Risk participants and none of the Low-Risk group ($p < 0.001$).

Violence and abuse

Reports of violence were higher in the High-Risk group. All participants in the High-Risk group reported experiencing physical violence compared to 91.4% in the Low-Risk group ($p < 0.001$). Sexual violence was reported by 24.6% of High-Risk participants and 5.1% of Low-Risk participants ($p < 0.001$). Emotional abuse was also significantly more common among High-Risk participants ($p < 0.001$).

Table 3 Collinearity diagnostics for predictors of risk cluster classification

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions
	1	2.221	1.000	0.03
	2	1.363	1.277	0.01
	3	0.321	2.631	0.01
	4	0.095	4.836	0.94

Cluster and bivariate analysis of risk profiles among AGYW in rural Uganda

Collinearity diagnostics and model stability

We conducted the collinearity diagnostics for the logistic regression model and revealed no significant multicollinearity concerns, as indicated by a Condition Index ranging from 1.000 to 4.836, well below the threshold of 10. The Eigenvalues ranged from 0.095 to 2.221, reflecting distinct contributions of predictors to the model. Variance proportions confirmed predictor independence, with “School Status” and “Sexual Activity” contributing predominantly to Dimension 4 (89% and 43%, respectively; Eigenvalue = 0.095), while “Transactional Sex” aligned with Dimension 3 (84%; Eigenvalue = 0.321). As shown in Table 3.

It is important to note that the very large odds ratios observed in the logistic regression model are partly attributable to the analytic design. Since participants were first categorized into High-Risk and Low-Risk groups using K-means clustering, subsequent regression predictors such as school enrollment, transactional sex, and sexual activity were already strongly aligned with these categories. This categorization inherently magnifies effect sizes, as the regression is essentially modeling predictors that were used in the formation of the outcome clusters. Sensitivity checks, including Variance Inflation Factor (VIF) analysis, confirmed that multicollinearity was not a concern. Therefore, the large odds ratios should be interpreted as reflecting the strong discriminating power of these predictors rather than model misspecification.

Logistic regression results following cluster and bivariate analysis of risk profiles among AGYW in rural Uganda

Building on the cluster and bivariate analysis, a binary logistic regression was conducted to identify predictors of High-Risk and Low-Risk classification among adolescent girls and young women as shown in Table 4. The model demonstrated excellent explanatory power, with a -2 Log Likelihood of 23,765.883, a Cox & Snell R Square of 60.4%, and a Nagelkerke R Square of 89.8%. These results indicate that the model accounts for nearly 90% of the variance in risk classification. As shown in Table 4, school enrollment, sexual activity, and transactional

Table 4 Adjusted odds ratios (AOR) and 95% confidence intervals for predictors of risk classification among AGYW

Variable	p-value	AOR (Exp(B))	95% CI for AOR
School Status	< 0.001	108.154	93.844–124.646
sexually activity	< 0.001	0.000	0.000–0.001
Transactional sex	< 0.001	1599.966	769.091–3328.465
Constant	< 0.001	5.370	

sex were statistically significant predictors of classification. School enrollment was associated with an adjusted odds ratio (AOR) of 108.154 (95% CI: 93.844–124.646, $p < 0.001$). Sexual activity was associated with an AOR of 0.000 (95% CI: 0.000–0.001, $p < 0.001$). Engagement in transactional sex was associated with an AOR of 1599.966 (95% CI: 769.091–3328.465, $p < 0.001$).

Discussion

The overarching objective of this study was to establish a comprehensive baseline risk profile of AGYW in rural Uganda, focusing on socio-economic, educational, and behavioral vulnerabilities. Specifically, we sought to (i) examine these vulnerabilities in detail, (ii) identify distinct risk profiles using cluster analysis, and (iii) determine the key predictors of high-risk behaviors such as transactional sex, school dropout, and exposure to violence. By doing so, we aimed to generate evidence that can inform targeted interventions and policy responses tailored to the needs of AGYW.

Our findings revealed that AGYW in rural Uganda face intersecting risks where poverty, education, violence, and behavioral factors combine to shape vulnerabilities. The clustering analysis showed clear disparities: those in the High-Risk category were more likely to engage in transactional sex, experience violence, and drop out of school. These patterns are not random but reflect broader pathways of risk rooted in socio-economic hardship, gender inequality, and lack of institutional support. This resonates with global literature stressing the need to address structural determinants of health to reduce risky behaviors among vulnerable populations [1].

Education emerged as a transformative protective factor. Girls enrolled in school were 108 times more likely to be classified as low-risk, a finding consistent with prior studies showing that education enhances agency, delays sexual debut, and reduces early marriage [22]. Nevertheless, persistent barriers poverty, early pregnancies, and entrenched cultural norms continue to limit access to schooling for many AGYW [21, 23]. Addressing these barriers requires scholarships, mentorship initiatives, and community-driven advocacy to ensure that education is both accessible and sustainable, particularly in rural Uganda.

Sexual activity and early sexual debut were strongly associated with high-risk classification. This underscores

the need for comprehensive sexual and reproductive health (SRH) education that contextualizes risk beyond biological knowledge, integrating discussions on gender, power, and economic vulnerability. Previous research affirms the effectiveness of peer-led counseling, contraceptive access, and cultural norm change in reducing early sexual initiation and risky behavior [6]. Integrated SRH programming under initiatives such as PEPFAR and DREAMS has demonstrated potential to empower AGYW, further reinforcing the importance of such approaches.

Transactional sex stood out as an acute marker of economic desperation. AGYW engaging in transactional sex were nearly 1600 times more likely to be in the High-Risk group, reflecting how poverty shapes unsafe survival strategies. This aligns with evidence that economic deprivation and lack of parental or household support drive AGYW into coercive or exploitative relationships [1, 9]. Addressing this pathway requires investment in economic empowerment programs, including vocational training, financial literacy, and microfinance initiatives, which can provide viable alternatives to exploitative dependency.

Violence was a pervasive issue across the High-Risk group. Many participants reported physical, sexual, and emotional abuse, which not only increased immediate risks but also contributed to long-term psychological harm such as anxiety, depression, and PTSD. Evidence from similar settings shows that violence disrupts educational opportunities and perpetuates cycles of exclusion and poverty [6, 21]. Protecting AGYW requires multi-sectoral strategies that include strengthening legal protections, community interventions to shift harmful norms, and survivor-centered services such as psychosocial support and safe spaces.

Substance use also emerged as both a symptom and a driver of risk. High-Risk AGYW were more likely to report alcohol and drug use, often as a coping mechanism for poverty or violence. However, substance use increased exposure to unsafe sexual behavior and violence. These findings echo prior research linking substance use to heightened vulnerability among adolescents [24]. Prevention and response programs particularly community-based peer networks and accessible counseling are critical to address both substance use and its underlying social causes.

Finally, the role of DREAMS in Uganda must be emphasized. As part of a broader initiative implemented in 10 high-burden HIV countries, DREAMS provides layered services safe spaces, para-social worker support, education subsidies, GBV prevention, and economic strengthening [20]. The baseline vulnerabilities we identified (transactional sex, school dropout, violence, and substance use) align with the program's focus areas. Our

findings therefore reinforce the relevance of DREAMS' approach and highlight how risk stratification at enrollment can be leveraged to target the most vulnerable girls more effectively.

In summary, this study shows that AGYW's vulnerabilities follow clear pathways: poverty drives transactional sex; transactional sex and violence increase exposure to exploitation; school dropout removes protective social networks; and substance use compounds risks. Interventions that simultaneously address these pathways through education, economic empowerment, GBV prevention, and psychosocial support are essential for building resilience and breaking cycles of risk among AGYW in rural Uganda.

Practical and policy implications

This study highlights the urgent need to support AGYW in rural Uganda by addressing the challenges they face in their daily lives. On a practical level, keeping AGYW in school is a powerful way to shield them from risks like early pregnancies, transactional sex, and violence. Efforts to reduce school dropouts could include scholarships for vulnerable families, community mentorship programs, and flexible re-entry policies for girls who leave school due to early motherhood. Empowering AGYW economically through vocational training, small business support, and savings programs can help them build independence and avoid exploitative relationships. Additionally, sexual and reproductive health education should not only be accessible but also relatable, addressing the specific needs and experiences of AGYW. Establishing safe spaces where girls can seek mental health support and assistance after experiencing violence or substance misuse is crucial for their recovery and resilience.

On a broader scale, policies must reflect a commitment to the holistic well-being of AGYW. Education policies should ensure that every girl has the opportunity to complete her studies, regardless of socio-economic barriers. Enforcing laws to prevent child marriages and gender-based violence while providing accessible and confidential reporting mechanisms can create safer communities for girls. Policies promoting reproductive health services need to account for cultural sensitivities while making care widely available and free from stigma. Economic programs must address the underlying poverty that drives many AGYW into high-risk behaviors by creating pathways to sustainable livelihoods for them and their families. A united effort involving government, civil society, and local leaders is needed to ensure these policies and interventions reach the girls who need them most. Together, we can create a future where AGYW have the resources and support to lead safe, healthy, and empowered lives.

Conclusion

This study highlights the compounded vulnerabilities faced by adolescent girls and young women (AGYW) in rural Uganda, where poverty, school dropout, violence, and transactional sex converge to heighten risk. While a large proportion were categorized as high-risk, education emerged as a strong protective factor, underscoring its role in resilience and empowerment. Conversely, economic desperation and exposure to violence deepened vulnerabilities, pointing to the need for interventions that go beyond HIV prevention to address broader socio-economic and gender-based challenges. These findings call for collective action by families, communities, policymakers, and partners to strengthen education, expand livelihood opportunities, and provide survivor-centered protection and support. Grounding programs like DREAMS in such baseline evidence will improve targeting and ensure more sustainable impact, ultimately creating safer and more enabling environments for AGYW to realize their potential.

Strengths, limitations, and directions for future research

Our study offers several strengths that enhance its reliability and impact. Using a combination of cluster analysis, bivariate analysis, and logistic regression, it provides a detailed understanding of the socio-economic, educational, and behavioral factors shaping the risk profiles of AGYW in rural Uganda. A large sample size of over 120,000 participants enhances statistical power and generalizability. The focus on key risk indicators, such as education, sexual activity, and transactional sex, alongside broader determinants like violence and substance use, ensures the findings are both actionable and contextually relevant for designing targeted interventions.

However, limitations exist. The cross-sectional design restricts causal inferences, necessitating longitudinal studies to track changes in risk factors over time. The reliance on self-reported data raises concerns about potential biases, particularly underreporting of sensitive issues like sexual activity and violence. Moreover, structural determinants, such as cultural norms or systemic barriers, are underexplored. Future research should include qualitative approaches to understand AGYW's lived experiences and contextual dynamics. Additionally, studies focusing on marginalized sub-groups, such as AGYW with disabilities or those in conflict zones, would enhance inclusivity. Addressing these gaps can guide more comprehensive strategies to support AGYW and inform inclusive interventions in similar contexts.

Another limitation concerns the unusually large odds ratios in the regression model. These reflect the strong separation produced by the prior K-means categorization of High-Risk and Low-Risk participants, which amplified

predictor effects. While diagnostic checks ($VIF < 2$) ruled out multicollinearity, the categorization step must be considered when interpreting effect sizes. As a methodological enhancement, future work will consider Latent Profile Analysis (LPA) to obtain model-based profiles and information-criterion guidance (AIC/BIC) on the number of classes. Comparing LPA with K-means would further test profile robustness and strengthen the validity of risk-group identification.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-025-25027-7>.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

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Authors' contributions

All authors contributed substantially to the conception, design, data collection, analysis, and manuscript preparation. Dedrix Stephenson Bindeeba: Conceptualization, Methodology, Data Curation, Formal Analysis, Writing – Original Draft, Supervision. Jane Senyondo Nakawesi: Investigation, Data Collection, Writing – Review & Editing. Andrew Mugisa: Statistical Analysis, Visualization, Writing – Review & Editing. Catherine Senyimba: Data Interpretation, Writing – Review & Editing. Ronald Mulebeke: Validation, Resources, Writing – Review & Editing. Yvonne Karamagi: Project Administration, Funding Acquisition, Writing – Review & Editing. All authors read and approved the final version of the manuscript.

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Data availability

The dataset used and analyzed in this study is available from the corresponding author upon reasonable request, subject to ethical approvals and confidentiality agreements.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. Ethical approval was obtained from the Mildmay Uganda Research and Ethics Committee (MUREC) (REC REF 0804–2018) and the Uganda National Council for Science and Technology (SS639ES). Since this study utilized secondary data, a waiver for informed consent was granted by the Mildmay Uganda Research Ethics Committee, as all data were fully anonymized before analysis.

Consent for publication

Not applicable. This manuscript does not contain any individual-level data (such as images, videos, or personal details) that require specific consent for publication.

Competing interests

The authors declare no competing interests.

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